

AEye

**Multimodal Analysis of Visual Behavior for Cheating Detection in Computer-Based
Exams**

A Thesis Project Presented to the Faculty of
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CHAPTER I

INTRODUCTION

Rationale of the Study

Cheating during examinations has become a growing concern as more schools and universities give tests on computers, both at home and inside the classroom. To address this, many institutions now use camera-based monitoring tools that watch a student's movements for signs of dishonesty. The difficulty is that these tools often treat ordinary, innocent movements as evidence of cheating. A student who glances away from the screen to think through a problem, looks down to write on scratch paper, turns their head while reading a long question, or moves their eyes while trying to recall an answer can all be marked as suspicious. The problem is worse for students whose bodies simply move differently, such as those with tremors, fidgeting, or other involuntary movements, and for those with facial features such as cross-eyed or squint-eyed conditions or eyeglasses that reflect light. In each of these cases the movement looks anomalous to the system even though no cheating is taking place. This raises a real concern about whether such tools can tell the difference between actual cheating and the normal range of movement found among different students.

Many studies have already looked at how to detect cheating during online exams, and they differ in how many signals they watch and how they capture them. Dilini et al. (2021) built a browser-based system that relies on eye-gaze tracking alone through a normal webcam, flagging the moments when a student's eyes move away from the screen. Other work adds head movement and head pose as further signals. Atoum et al. (2017) presented an early system that continuously estimated six behavior cues, user verification, text detection, voice detection, active window detection, gaze estimation, and phone detection, using a webcam, a wearable camera, and a microphone, and combined them over a time window to decide

whether cheating was happening. More recent studies add deep learning across several signals: Al-Mukhtar et al. (2024) combined face detection, eye tracking, and head movement and reported about 87 percent accuracy, while Naveen et al. (2025), in the AutoOEP framework, integrated facial cues, head pose, gaze, and hand-object interaction using a dual-camera setup, one camera facing the student and one viewing the workspace, together with a time-based model, and reported about 90.7 percent accuracy. A review by Khalil et al. (2025) likewise found that tools mixing face, head, and eye signals outperformed those relying on a single cue. Yet across this body of work, many widely used tools still lean on a single signal, usually eye gaze, and on a single front-facing view, treating almost any look away from the screen as suspicious and leaving the desk and the student's hands outside the camera's view.

These same studies provide evidence that three gaps remain. First, single-signal, single-view designs are demonstrably error-prone: Dilini et al. (2021) acknowledge that their own eye-only system cannot account for students with special needs, and Naveen et al. (2025) report that single-cue, single-frame analysis produces a high rate of false positives. A front-only view is a particular weakness, since it cannot see the desk surface or the student's hands, so it cannot tell whether a downward glance is a student writing a solution on scratch paper or reaching for a hidden device. Second, almost all of these tools compare every student against one fixed standard of "normal," and research shows this is unfair to students whose bodies move differently. Njeru et al. (2025), interviewing students who use disability accommodations, found that involuntary movements and accommodation use produced repeated false flags, in one case ending a student's exam after too many were triggered, and Coghlan et al. (2021) warn that neurodivergent students and students with disabilities are red-flagged through the ordinary manifestation of their own behavior. The same problem applies to facial constraints: students who wear eyeglasses, or who have cross-eyed or squint-eyed conditions, present a face the system was not built to read and are penalized for features they cannot change. Third,

most existing tools are built and tested in clean, well-lit conditions, not in the ordinary home and classroom setups common in Philippine schools. Ye et al. (2022) showed that the accuracy of widely used pose- and eye-landmark models falls measurably as image resolution, color depth, and lighting drop, while Ansari et al. (2023) found that a person-specific model calibrated to the individual user clearly outperforms a universal one. Together, these gaps mean that current tools may catch real cheating but also punish honest students whose only fault is the way their body naturally moves or the ordinary act of writing on paper the camera cannot see.

The study aims to address these gaps by building AEye, a standalone desktop application that works alongside the school's e-class, which is where examinations are hosted. AEye is used in two modes, a student mode for taking the examination and a proctor mode for monitoring it, and is backed by a central server. The study intends to include several elements that current single-signal, single-view tools leave out. First, students sign in to AEye with their own login using their school-issued student ID, and then take their e-class examination inside AEye's controlled window, which displays the e-class exam page within a locked environment. Second, because AEye takes the place of a lockdown browser, its student mode restricts the exam environment, running in fullscreen and limiting switching to other applications during the examination. Third, instead of relying on the eyes alone, the study includes multiple cues together, eye gaze, head pose, and facial expression, as the basis for judging behavior. Fourth, it reads these cues from two cameras rather than one: a front-facing camera at the student's line of sight that captures the face for eye gaze, facial landmarks, and head pose, and an elevated side camera that captures the upper body, so that posture and arm and hand movement, such as reaching toward or under the desk, are also visible. Fifth, it includes a short calibration before each exam that records two areas of interest for that student's setup, the screen (Zone 1) and the desk (Zone 2), so that looking at the screen or looking down to write on paper is treated as legitimate rather than suspicious, and that also accounts for the examination environment and

the student's personal facial baseline, including features such as eyeglasses or cross-eyed and squint-eyed conditions. Sixth, it includes a per-student behavioral record that accounts for each student's own normal behavior, including involuntary movements such as tremors or fidgeting, so that what counts as unusual is considered against that student's own baseline rather than a generic standard. Seventh, it includes detection based on patterns of behavior over time rather than a single moment. Finally, the study includes a confidence score with a severity level, delivered in real time to a human proctor in the application's proctor mode together with time-stamped evidence, with the proctor making the final decision. By bringing these elements together, the study aims to be more accurate and fairer than current single-signal tools, especially in the real conditions of Philippine schools.

Review of Related Works

This section discusses existing studies and systems related to automated exam proctoring. It is organized by topic and ends by comparing existing works to the proposed system. The studies cover online exam monitoring in general, eye-only detection and its weaknesses, multimodal and multi-camera detection, the MediaPipe tool, personal calibration and areas of interest, time-based behavior models, the Philippine setting, and privacy and fairness concerns.

Online Exam Monitoring and Cheating

As more schools give tests on computers, keeping exams honest has become a big concern, and this grew even more during the COVID-19 pandemic. Malhotra and Chhabra (2026) looked at eighty past studies and found that having a person watch every student is hard to do for large classes, which is why more schools now use software that uses machine learning

to spot cheating. Still, people do not fully agree on how well these tools work. Gribbins and Bonk (2023) asked 158 teachers about exam-monitoring software, and while some said it helped, others worried about student privacy and the stress it caused. This shows that a good system must not only catch cheating but also avoid hurting honest students with wrong alarms.

Eye-Gaze Detection and Why It Falls Short

Where a student looks is one of the most common signals used to spot cheating. Dilini et al. (2021) built a tool that tracks eye movement using a normal webcam to flag odd behavior, noting that eye movement can even show when a student is trying to recall an answer. But a system that watches the eyes only often gets things wrong. A later multimodal study pointed out that eye-only tools treat any look away from the screen as a sign of cheating, which is too simple (Naveen et al., 2025). A student who looks away to think, writes on scratch paper, or wears glasses that reflect light may set off a false alarm. This is why many studies now mix several signals instead of one.

Using More Than One Signal

Many studies agree that using several signals together works better than using just one, and recent work also shows that a single front-facing view is not enough. The early system by Atoum et al. (2017) used six signals at once and checked them over short time windows, while also saying their tool was not perfect, so cases that looked like cheating should still be checked by a person. Newer tools add deep learning. Al-Mukhtar et al. (2024) put face detection, eye tracking, and head movement into one model and reported about 87 percent accuracy. A review by Khalil et al. (2025) also found that tools mixing face, head, and eye tracking did better than tools that used one signal alone. Going further, Naveen et al. (2025) in AutoOEP used a dual-camera setup, a front camera for the face and a side camera for the workspace, so that the system could watch not only the face but also the hands and the desk, tracking how close a student's hands came to prohibited items; the authors note that a front-only view leaves the

workspace as a blind spot. These studies back the idea of mixing eye gaze, head pose, and facial expression, and of adding a second camera that can see the desk and hands, so that ordinary actions like writing on paper are not mistaken for cheating.

MediaPipe for Reading the Face and Head

To read the face on a normal computer, a light and fast tool is needed. Google (2025) explains that MediaPipe can find 478 points on the face, give scores for facial expressions, and work out head position, all in real time using just one camera and no special hardware. Nguyen et al. (2025) used MediaPipe to pull out face and body features, fed them into a model, and reached over 85 percent accuracy in spotting seven kinds of cheating. Hisey et al. (2025) also used MediaPipe to track eyes inside a web browser. This supports using MediaPipe as the main tool for reading the face, even with cheap webcams.

Personal Calibration for Each Student

Tools that compare everyone to one fixed standard do poorly when faces, cameras, and lighting are not the same. Cheng et al. (2024) explain that a short personal setup makes results more accurate for new users and new settings, and Park et al. (2019) found that even a few setup samples lower the error. This helps the most when the camera is poor. Ansari et al. (2023) trained models on low-quality webcam images and found that a quick setup step improved accuracy without needing any extra hardware. This supports the present study's plan to use a single thirty-second calibration at the start of each exam, which checks the environment and the state of the student's face and also records the student's areas of interest, the screen and the desk, so the system knows where it is normal for that student to look and where their hands normally rest.

Watching Behavior Over Time

Recent work shows that cheating is easier to spot as a pattern over time than in a single frozen moment. The AutoOEP study (Naveen et al., 2025) reached about 90.7 percent accuracy using a time-based model that reads a sequence of movements rather than a single frame, which the authors note reduces the false positives common in frame-by-frame analysis. Sani et al. (2023) also reached about 90 percent accuracy with a similar time-based model. At the same time, the basic time-window method of Atoum et al. (2017) still works when there are fewer resources. This matches the present study's plan, which counts repeating patterns over time and feeds them into a confidence score, while keeping a record of each student so that odd behavior is judged against that student's own habits.

Low-Resource and Philippine Settings

Filipino students take exams in conditions that are far from the clean setups most tools are tested in. Rotas and Cahapay (2020) found that students dealt with weak internet, few learning materials, power cuts, and poor study spaces. Fabito et al. (2021) found that for computing students, the lack of a proper study area was one of the top problems, ranking even higher than poor internet. Barrot et al. (2021) likewise found that the home study setup was the biggest problem students faced. These conditions, uneven lighting, cramped desks, and varied equipment, are exactly where generic proctoring tools tend to misread ordinary behavior, which is why the present study is designed and evaluated for these real settings rather than for an ideal laboratory. Rather than assuming a single perfect webcam, AEye is built to use two ordinary cameras, such as a laptop webcam paired with a phone camera, a deliberate choice that trades a small amount of extra setup for a clearer view of both the face and the workspace.

Privacy, Fairness, and Ethics

Because these systems collect data about a person's face and movement, privacy and fairness matter a great deal. Terpstra et al. (2023) found that what students accept depends on the situation, and that students worry when their data is used for something other than what it was meant for. Njeru et al. (2025) report that exam-monitoring tools may unfairly flag students with darker skin or students who move in different ways, such as those who are neurodivergent or who have involuntary movements. This is exactly the kind of unfairness a per-student baseline is meant to reduce, since each student is judged against their own normal rather than a fixed standard. Coghlan et al. (2021) add that wrong alarms are unfair to honest students.

Comparison to the Proposed System

Compared to existing works, the proposed system, AEye, brings several ideas together that are usually treated apart. Existing tools often use only one signal (eye gaze), while AEye uses three (eye gaze, head pose, facial expression). Most tools rely on a single front-facing camera, while AEye uses two, a front camera at the student's line of sight for the face and an elevated side camera for the upper body, so that posture and hand movement toward or under the desk are visible; this follows the dual-camera direction of Naveen et al. (2025) but pairs it with a per-student baseline. AEye also defines two areas of interest during calibration, the screen and the desk, so that looking down to write on paper is treated as legitimate rather than suspicious. Existing tools usually compare every student to one fixed standard, while AEye keeps a personal record for each student and judges them against their own normal, which is fairer to students with involuntary movements. Most tools judge a single frame, while AEye counts patterns over time. Most tools are tested in clean setups, while AEye is designed for the uneven lighting and varied equipment common in Philippine schools. Finally, where other systems leave results in logs for later review, AEye pushes flagged incidents in real time to the proctor mode of the application, where a proctor monitoring several students can act at once. Identity, which other systems handle through separate face recognition, is handled here by

AEye's own login using the school-issued student ID that the student enters when signing in, while the examination itself is taken inside AEye's controlled window on the school's e-class.

Project Objective

General Objective

To develop AEye, a standalone desktop application that works alongside the school's e-class, backed by a central server and used in student and proctor modes, that displays the e-class examination inside a controlled window while examining eye gaze, head pose, and facial expression from a front-facing camera together with upper-body posture and hand movement from an elevated side camera, defines the screen and desk as areas of interest through a short calibration, learns each student's normal behavior over time, and delivers a confidence score in real time to human proctors, in order to help keep computer-based examinations honest in schools with limited resources.

Specific Objectives

1. To design and build a standalone desktop application with a student mode and a proctor mode, in which the student mode displays the e-class examination inside a controlled window, reads and processes a student's eye gaze, head pose, and facial expression from a front-facing line-of-sight camera, and upper-body posture and arm and hand movement from an elevated side camera, as the examination takes place,

while restricting the exam environment by running in fullscreen and limiting switching to other applications.

2. To implement a single thirty-second calibration at the start of each examination that records the student's areas of interest (Zone 1, the screen, and Zone 2, the desk) and checks both the environment (camera quality and lighting) and the state of the student's face (such as eyeglasses, cross-eyed or squint-eyed conditions, and other features that affect how the face appears on camera), giving the system a personal reference even during the student's first examination.
3. To develop a model that combines the signals from both cameras (eye gaze, head pose, and facial expression from the front camera, and upper-body posture and hand movement from the side camera) into a single overall judgment of how suspicious a student's behavior is, making the system more reliable than tools that rely on one signal or a single view.
4. To develop a pattern-based detection method that examines a student's movements over time rather than a single moment, counting repeating patterns such as the look-away, read, and return cycle typical of cheating, and feeding these counts into a confidence score with a severity level.
5. To add a feature that learns each student's normal exam behavior across several examinations and remembers it, so that involuntary movements such as tremors or fidgeting are treated as normal for that student and not as cheating.
6. To add a feature in the proctor mode that flags suspicious moments as they happen and delivers them in real time from the central server, saving clear, time-stamped evidence such as screenshots tied to the student's school ID, so that a human proctor monitoring several students at once can review each incident and make the final decision.

7. To evaluate how well the system performs under different camera qualities and lighting conditions by measuring how accurate it is and how much wrong accusations decrease after it has learned each student, while ensuring that consent, data security, and student privacy are handled properly.

Project Scope and Limitation

Scope

The system, AEye, is a standalone desktop application for Windows, backed by a central server, that works alongside the school's e-class, which hosts the examinations. Students sign in to AEye using their school-issued student ID, then take their e-class examination inside AEye's controlled window, which displays the e-class exam page within a locked environment. AEye is used in two modes. In student mode, running on the student's Windows computer, it reads the student's eye gaze, head pose, and facial expression in real time from a front-facing camera placed at the student's line of sight, and the student's upper-body posture and arm and hand movement from an elevated camera placed to the side; it also restricts the exam environment by running in fullscreen and limiting switching to other applications during the examination. Before each exam, it runs a single thirty-second calibration that records the student's areas of interest (the screen and the desk) and checks the environment and the state of the student's face. During the exam, it counts repeating patterns of behavior over time, judged against the student's own baseline, and produces a confidence score with a severity level. Suspicious moments are flagged and sent through the central server to the proctor mode, where a proctor monitoring several students can see time-stamped screenshots tied to each student's student ID and make the final decision. The system is designed for both home and classroom examinations in the ordinary lighting and equipment conditions common in Philippine schools.

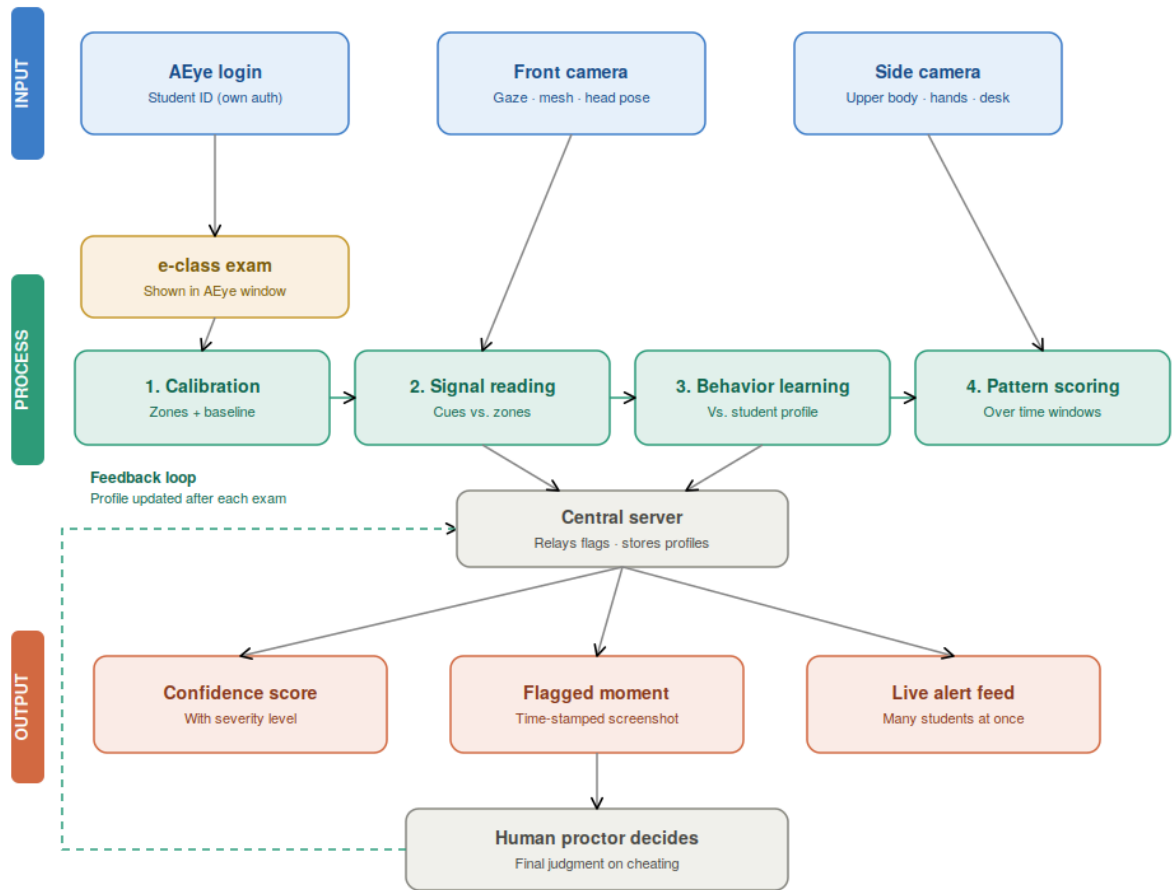
Limitations

1. AEye does not perform face recognition or biometric authentication. Identity is taken from the application's own login, where the student signs in with their school-issued student ID, so the system trusts that the logged-in account is the correct student.
2. AEye does not make final judgments on cheating. It only flags suspicious moments and provides confidence scores and screenshots; the final decision is always made by a human proctor.
3. Because the system is designed to be fair to students with involuntary movements such as tremors, the design has not been clinically tested with medical patient groups. The fairness benefit comes from the per-student baseline and is shown by general accuracy and false-positive measurements.
4. The exact display format of the confidence score and severity level, including the numeric range and how patterns are shown to the proctor, will be finalized during development based on usability testing with proctors.
5. The system requires two working cameras, a front-facing camera and an elevated side camera. It cannot read behavior from a covered, broken, or disconnected camera, and if one camera fails the corresponding signals are unavailable; very dark rooms may also reduce accuracy even after calibration.
6. The system does not detect cheating that does not involve visible behavior, such as memorized notes, hidden earpieces with silent receivers, or collusion outside the camera's view.

7. AEye works alongside the school's e-class, which hosts the examinations, but it does not integrate with the e-class at the account or session level; students authenticate through AEye's own login, and AEye displays the e-class exam page inside its controlled window rather than exchanging login sessions with the e-class server. Because there is no session-level integration, AEye does not by itself force an examination to be taken through it; as with a lockdown browser, the instructor or proctor requires students to use AEye. The application is built for the Windows operating system; other operating systems such as macOS, Linux, ChromeOS, and mobile platforms are not supported and are outside the scope of this study. The student mode must be installed on each examination computer, and its environment restrictions apply within Windows and are not a guarantee against every possible workaround. For this study, AEye is evaluated against a researcher-controlled test instance of the e-class platform; deployment on the university's live e-class would require configuration and cooperation from the e-class administrator, which is outside the scope of this study.

Conceptual Framework

The conceptual framework of AEye follows an Input-Process-Output flow. The framework shows how raw information from the student's session, captured by two cameras in the application's student mode while the student takes the e-class examination inside AEye's controlled window, is processed on the student's computer, relayed through a central server, and turned into a confidence score that a proctor can review in real time in the application's proctor mode.



Input

The system takes several kinds of input. First, the student's student ID is taken from AEye's own login, where the student signs in before the examination. Second, a live video feed from a front-facing camera at the student's line of sight captures the face for eye gaze, facial landmarks, and head pose. Third, a live video feed from an elevated side camera captures the upper body, including posture and arm and hand movement toward or under the desk. Fourth, the student's areas of interest (the screen and the desk) together with environment and lighting information are captured during the thirty-second calibration at the start of each exam. The

examination content itself is served by the school's e-class and displayed inside AEye's controlled window.

Process

The process has four steps. In the first step, calibration, the system uses MediaPipe to record the student's normal face state (including eyeglasses or other features), the two areas of interest (screen and desk), and the environment (camera quality and lighting). In the second step, signal reading, MediaPipe reads eye gaze, head pose, and facial expression from the front camera, while the side camera provides upper-body posture and hand movement, and the system checks where the student is looking and where the hands are relative to the areas of interest. In the third step, behavior learning, the system compares the current signals to the student's saved behavioral profile from past exams, so that involuntary movements and personal habits, and legitimate actions such as writing on paper within the desk area, are treated as normal. In the fourth step, pattern detection and scoring, the system counts repeating patterns of behavior over short time windows and feeds them into a sequence-based model with a simpler time-window rule as a backup, producing a confidence score with a severity level.

Output

The output is a set of flagged moments, each with a confidence score, a severity level, the pattern that triggered the flag, and a time-stamped screenshot. All outputs are tied to the student's school ID and sent through the central server to the proctor mode, where a human proctor monitoring several students sees the flagged moments and makes the final decision on whether cheating took place. The flagged moments are also saved for later review. Over time, the student's behavioral profile is updated and stored on the central server, so that future exams build on what the system has already learned.

Feedback Loop

Each completed exam updates the student's behavioral profile. The next time the same student takes an exam, the system has more information about what is normal for them, which is expected to reduce wrong alarms in later sessions. This feedback loop is one of the main reasons the system is expected to be fairer to students whose normal behavior involves more movement.

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